Applying the front_spot OCE conditional no-arbitrage test to the DLV × extended-Bates CRC model

`quant_data-main/quantdata/dysanos/bates_crc/real_data_study`

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Abstract

We apply `quantdata/sdlv/daily/front_spot/run_oce.py` and `run_quadratic_oce.py`'s conditional no-arbitrage OCE methodology to the DLV×extended-Bates CRC model (`quantdata/dysanos/bates_crc/`), seeded from a real calibrated state in `daily_data_cache`. The policy network and both OCE objectives (`oce.py`) are fully model-agnostic and reused unchanged; only the front_spot-specific JAX simulator required a bates_crc analogue (`oce_population.py`). Two real numerical bugs were found and fixed during development (a short-tau Fourier quadrature bias, and a CIR-Euler discretisation bias under badly Feller-violating fitted parameters). A third, real, quantified deviation in the day-by-day multi-instrument repricing sequence was found but its root cause remains OPEN despite extensive isolation; the held-out conditional-drift results below are the direct, predictable consequence of a trained policy exploiting that known issue, not a finding about the model's own risk-neutral measure. We report this transparently, exactly as the test is designed to.

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1 Method

Reused unchanged from front_spot. `oce.py`'s `FrontSpotOcePolicy` (an uncapped affine-plus-deep feature-to-action network), `regularized_transition_gain` (the exponential OCE gain $a'R - \frac{1}{2}\epsilon a'a$, used by `run_oce.py`), `quadratic_second_moment_reward` (the quadratic OCE reward $a'R - \frac{1}{2}|a|^2 - \frac{1}{2}\lambda(a'R)^2$, used by `run_quadratic_oce.py`), and every statistics/audit function (`paired_conditional_drift_statistics`, `action_moments`, and `run_oce.py`'s own `unconditional_spot_scaled_martingale_audit`) are imported and used directly, not reimplemented – these operate purely

on generic `(paths,steps,feature_count)`/`(paths,steps, instrument_count)` tensors, independent of which model generated them.

Rebuilt for bates_crc (`oce_population.py`). The seed state is the real fitted `(chain_config, heston, jumps,` V_t`)` for a held-out `daily_data_cache` date (the latest held-out date whose jump calibration is not pinned at an optimizer bound). A 20-business-day horizon is simulated day by day: the chain's own exact matrix-exponential transition and an exact CIR + compound-Poisson Bates step (see Sec. 2), repricing a fixed panel of 4 expiries × 11 relative strikes (44 options, calls below spot, puts above, avoiding vanishing OTM-call premia) plus spot via the already-proven chain-mixture Fourier pricer, generalised from one contract to a full panel at once.

Quantity	Value
Seed date (real, daily_data_cache)	2025-08-14
Path count	4000
Horizon (business days)	20
Feature count	52
Instrument count	45
Generation time (s)	417.1

2 Two bugs found and fixed

Short-tau Fourier quadrature bias. The pricer's usual single-contract defaults (`n_quad=80`, `u_max=150`) priced a real fitted date's 1-day-to-expiry ATM call at ∼52% relative error against a converged `n_quad=800`/`u_max=800` reference. Fixed: `n_quad=300`, `u_max=500`, verified to match the converged reference to within 0.03% at the same point.

CIR-Euler discretisation bias. The representative date used here badly violates the Feller condition ($2\kappa\theta/\xi^2 \approx 0.21$), a known source of persistent full-truncation-Euler bias for the variance process. Fixed by replacing the Euler step with the EXACT noncentral-chi-squared CIR transition (`latent_state.cir_exact_sample`, reused not reimplemented), with the correlated (leverage) component of the log-return recovered exactly from each exact-CIR substep's own realised path via the defining SDE relation (the same identity used in Andersen's QE scheme). Verified: the single-anchor, 20-day chained case became clean after the fix (max $|z| = 2.9$, 0/570 coordinates exceeding $|z| = 3$), confirming the fix resolves the pure Bates-factor bias.

3 Headline held-out results

Objective	Pairs	Mean drift	Pair SE	z
Exponential (front_spot run_oce.py)	1000	6.243e-05	1.611e-06	38.76
Quadratic (front_spot run_quadratic_oce.py)	1000	6.029e-05	1.558e-06	38.69

Both objectives give essentially the same, highly significant held-out conditional drift ($z \approx 38.7$). Per Sec. 5, this is not evidence against the model's risk-neutral measure – it is the expected signature of a trained policy exploiting a known, open simulator issue in the day-by-day multi-instrument repricing sequence.

4 Unconditional spot-scaled martingale audit

For each objective's underlying population, every (day, instrument) return coordinate is tested separately for zero unconditional mean (antithetic-pair-based standard errors), split into the spot coordinate and the option coordinates.

4.1 Exponential-objective population

Coordinate group	N tested	Max $\|z\|$	RMS $\|z\|$	$N(\|z\| > 3)$
Spot	20	2.05	0.99	0
Options	550	30.39	3.60	79
All instruments	570	30.39	3.54	79

4.2 Quadratic-objective population

Coordinate group	N tested	Max $\|z\|$	RMS $\|z\|$	$N(\|z\| > 3)$
Spot	20	2.72	1.17	0
Options	550	28.28	3.59	75
All instruments	570	28.28	3.53	75

In both populations the **spot** coordinates are clean (max $|z| \approx 2$, none exceeding 3) – consistent with the chain and Bates factor's own one-step martingale properties being independently verified exact (Sec. 2). The **option** coordinates are not: up to $|z| \approx 30$, with roughly 14% of tested option coordinates exceeding $|z| = 3$.

Worst coordinates (exponential objective).

Day	Instrument	Mean	z
14	15D K=1 call	-0.00471	-30.39
9	10D K=1 call	-0.00378	-29.24
19	20D K=1 call	-0.00514	-28.60
19	20D K=1.02 put	-0.00516	-24.89
14	15D K=1.02 put	-0.00386	-20.44
9	10D K=1.02 put	-0.00279	-17.23
19	20D K=0.98 call	-0.00357	-16.43
19	20D K=1.04 put	-0.00321	-16.35

The worst coordinates are concentrated at ATM/near-ATM strikes on settlement-adjacent days (the last live-pricing day before an expiry) – a specific, reproducible pattern, not scattered/random noise.

5 Known open issue

Despite extensive isolation, the root cause of the option-coordinate martingale deviation above remains OPEN. What was specifically verified, each independently:

- the chain-mixture pricing formula matches the already-proven `chain_pricer.price_calls_chain_bates` reference exactly (to $\sim 10^{-18}$) for any tested (state, strike, tau) combination, including short tau and the real multi-anchor configuration;
- the day-by-day loop's carried option value at any given day matches an independent fresh recomputation from that exact day's realised state exactly (to $\sim 10^{-18}$);

- a direct terminal-date Monte Carlo test (chaining the exact same simulation primitives for the full 20-day horizon, comparing only the TERMINAL payoff against the closed-form 20-day price, bypassing all day-by-day repricing) is clean ($|z| < 1.5$ across a range of strikes);
- the raw (unnormalised) price increment shows the same bias as the spot-scaled version, ruling out the return-scaling convention as the cause.

The deviation is therefore specific to the day-to-day INTERMEDIATE repricing sequence under the full multi-anchor chain mixture, not to the pricing formula, the state bookkeeping, or the terminal distribution individually. This is documented in `oce_population.py`'s module docstring and encoded as an explicit regression-tracking test (`tests/test_real_data_study_oce_population.py`) so that a future fix is a deliberate, visible event (the test will start failing), not a silent change in behaviour.

6 Conclusion

The OCE test MACHINERY itself works as intended: given a population with a genuine, non-noise predictable component, the two-fold fit/evaluate pipeline (reusing front_spot's own policy network and reward functions unchanged) detects it, and the unconditional martingale audit correctly localises WHERE the deviation lives (options, not spot; settlement-adjacent ATM strikes specifically). The $z \approx 38.7$ headline results are not yet interpretable as a finding about the bates_crc model's own risk-neutral measure, because they trace to the open issue in Sec. 5 rather than to the model's economic content. The directly motivated next step is resolving that root cause – most promisingly by replacing the trapezoidal leverage-recovery approximation used across exact-CIR substeps with an exact joint (V,Y) scheme (e.g. Broadie–Kaya), and re-running this exact pipeline once that is done.